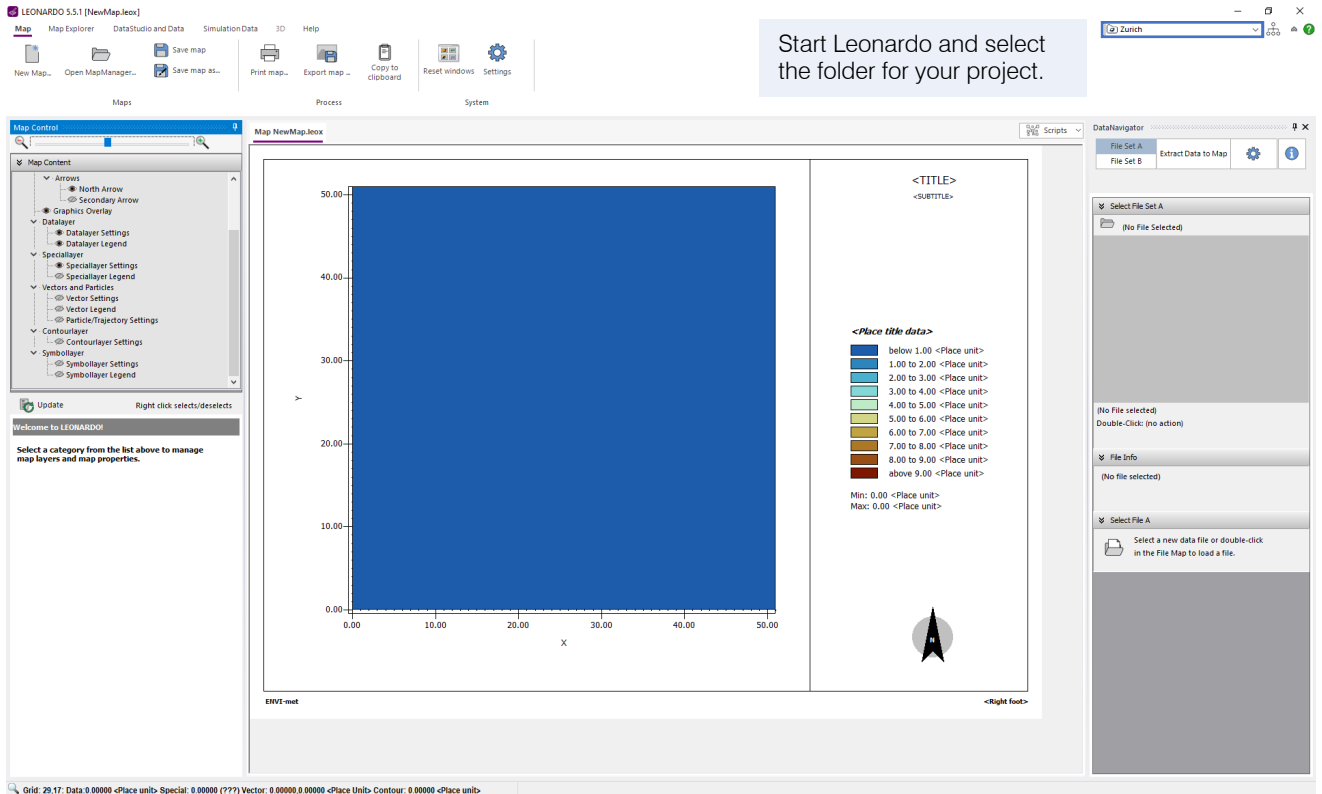
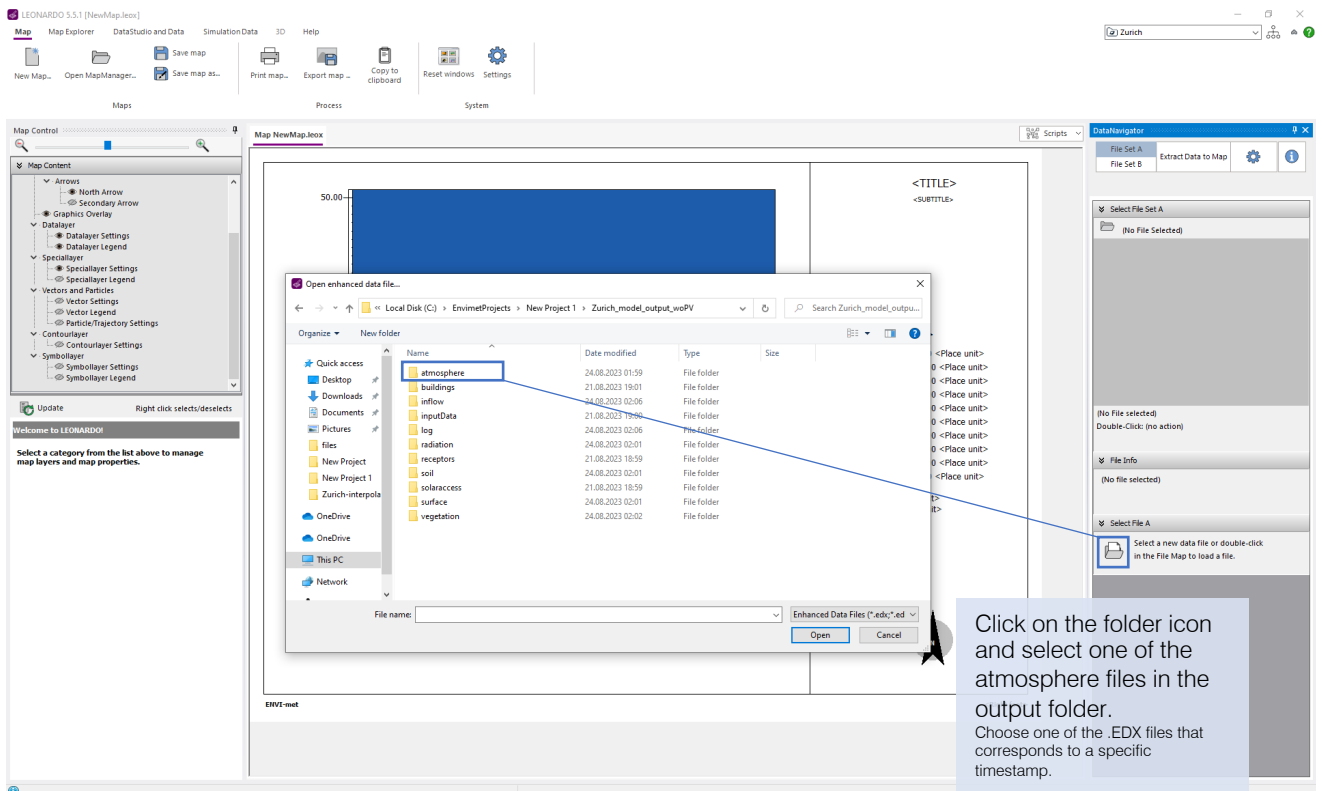


Visualization with Leonardo

1



2



Visualization with Leonardo

3

LEONARDO 5.5.1 [NewMap.jscx]

Map Explorer DataStudio and Data Simulation Data 3D Help

New Map... Open MapManager... Save map... Save map as...

Print map... Export map... Copy to clipboard

Reset windows Settings

Maps Process System

Map Control

Map Content

- Arrows
 - North Arrow
 - Secondary Arrow
- Graphics Overlay
- Datalayer
 - Datalayer Settings
 - Datalayer Legend
- Speciallayer
 - Speciallayer Settings
 - Speciallayer Legend
- Vectors and Particles
 - Vector Settings
 - Vector Legend
 - Particle/Trajectory Settings
- Contourlayer
 - Contourlayer Settings
- Symbollayer
 - Symbollayer Settings
 - Symbollayer Legend

Update Right click selects/deselects

Welcome to LEONARDO!

Select a category from the list above to manage map layers and map properties.

Map NewMap.jscx

50.00
40.00
30.00
20.00
10.00
0.00

0.00 10.00 20.00 30.00 40.00 50.00

X

Y

<TITLE>
<SUBTITLE>

<Place title data>

- below 1.00 <Place unit>
- 1.00 to 2.00 <Place unit>
- 2.00 to 3.00 <Place unit>
- 3.00 to 4.00 <Place unit>
- 4.00 to 5.00 <Place unit>
- 5.00 to 6.00 <Place unit>
- 6.00 to 7.00 <Place unit>
- 7.00 to 8.00 <Place unit>
- 8.00 to 9.00 <Place unit>
- above 9.00 <Place unit>

Min: 0.00 <Place unit>
Max: 0.00 <Place unit>

ENVI-met

<Right foot>

Grid: 21.10 Data: 0.00000 <Place unit> Special: 0.00000 (???) Vector: 0.00000, 0.00000 <Place Unit> Contour: 0.00000 <Place unit>

DataNavigator

File Set A Extract Data to Map File Set B

File map Set A

C:\Zurich_model_output_wop\atmosphere

24.07.2022

00:00 03:00 06:00 09:00 12:00 15:00 18:00 21:00 24:00

25.07.2022

00:00 03:00 06:00 09:00 12:00 15:00 18:00 21:00 24:00

AltstettenLinden_AT_2022-07-24_12.00.01.EDX

Double-Click (no action)

Data File A

AltstettenLinden_AT_2022-07-24_12.00.01.EDT

230 x 230 x 40 grids

File type: Atmosphere Data

Select Main Data File A

Select a new data file or double-click in the File Map to load a file.

You will find the time span of the simulation in this section. Please select one of the timestamps.

4

LEONARDO 5.5.1 [NewMap.jscx]

Map Explorer DataStudio and Data Simulation Data 3D Help

New Map... Open MapManager... Save map... Save map as...

Print map... Export map... Copy to clipboard

Reset windows Settings

Maps Process System

Map Control

Map Content

- Arrows
 - North Arrow
 - Secondary Arrow
- Graphics Overlay
- Datalayer
 - Datalayer Settings
 - Datalayer Legend
- Speciallayer
 - Speciallayer Settings
 - Speciallayer Legend
- Vectors and Particles
 - Vector Settings
 - Vector Legend
 - Particle/Trajectory Settings
- Contourlayer
 - Contourlayer Settings
- Symbollayer
 - Symbollayer Settings
 - Symbollayer Legend

Update Right click selects/deselects

Welcome to LEONARDO!

Select a category from the list above to manage map layers and map properties.

Map NewMap.jscx

330.00
300.00
270.00
240.00
210.00
180.00
150.00
120.00
90.00
60.00
30.00
0.00

0.00 30.00 60.00 90.00 120.00 150.00 180.00 210.00 240.00 270.00 300.00 330.00

X (m)

Y (m)

Figure 1: AltstettenLinden
12.00.01 24.07.2022
x/y Cut at k=4 (z=1.3500 m)

Potential Air Temperature

Min: 30.32 °C
Max: 34.43 °C

ENVI-met

<Right foot>

Click on "Extract Data to Map"

DataNavigator

File Set A Extract Data to Map File Set B

Data Source

Source: File Set A File Set B

Assign Map Layers and Elements

Data: Potential Air Temperature (°C)

Contour: <unchanged>

Symbol: <unchanged>

Vector: X: Flow u (m/s) Y: Flow v (m/s) Z: Flow w (m/s)

Special: Objects (1)

Arrows: Primary arrow: Import model ration Secondary arrow: <nothing>

Extract 2D Map

Type of 2D view: X-Y X-Z Y-Z

Handle topography: (No topography in file)

Position of view plane: k=4 z=1.3500 meter

Extract data from file

Extract 2D: AltstettenLinden_AT_2022-07-24_12.00.01.EDT

Reference: Observation

Swap Ref-Obs

Extract 3D Map

First, select the data type. Then, set the position of the view pane. Lastly, click on "Extract 2D"

Visualization with Leonardo

5

LEONARDO 5.5.1 [NewMapJeeX]

Map Explorer DataStudio and Data Simulation Data 3D Help

New Map... Open MapManager... Save map as... Print map... Export map... Copy to clipboard... Reset windows Settings

Maps Process System

Map Control

- Arrows
 - North Arrow
 - Secondary Arrow
- Graphics Overlay
- Datalayer
 - Datalayer Settings
 - Datalayer Legend
- Speciallayer
 - Speciallayer Settings
 - Speciallayer Legend
- Vectors and Particles
 - Vector Settings
 - Vector Legend
- Particle/Trajectory Settings
- Contourlayer
 - Contourlayer Settings
 - Contourlayer Legend
- Symbollayer
 - Symbollayer Settings
 - Symbollayer Legend

Update Right click selects/deselects

Welcome to LEONARDO!

Select a category from the list above to manage map layers and map properties.

Map NewMapJeeX

Figure 1: AltstettenLinden
12.00.01 24.07.2022
x/y Cut at k=4 (z=1.3500 m)

Potential Air Temperature

- below 30.73 °C
- 30.73 to 31.14 °C
- 31.14 to 31.56 °C
- 31.56 to 31.97 °C
- 31.97 to 32.38 °C
- 32.38 to 32.79 °C
- 32.79 to 33.20 °C
- 33.20 to 33.61 °C
- 33.61 to 34.02 °C
- above 34.02 °C

Min: 30.32 °C
Max: 34.43 °C

Flow v

- 2.00 m/s
- 4.00 m/s
- 6.00 m/s
- 8.00 m/s
- 10.00 m/s

Right-click on 'Vector Settings' and 'Legend' to turn the layers on.

Grid: 40.106 Data: No Data °C Special: 1.00000 (Buildings) Vector: 0.00000,0.00000 m/s Contour: 0.00000 °Place units

6

LEONARDO 5.5.1 [NewMapJeeX]

Map Explorer DataStudio and Data Simulation Data 3D Help

New Map... Open MapManager... Save map as... Print map... Export map... Copy to clipboard... Reset windows Settings

Maps Process System

Map Control

- Arrows
 - North Arrow
 - Secondary Arrow
- Graphics Overlay
- Datalayer
 - Datalayer Settings
 - Datalayer Legend
- Speciallayer
 - Speciallayer Settings
 - Speciallayer Legend
- Vectors and Particles
 - Vector Settings
 - Vector Legend
- Particle/Trajectory Settings
- Contourlayer
 - Contourlayer Settings
 - Contourlayer Legend
- Symbollayer
 - Symbollayer Settings
 - Symbollayer Legend

Update Right click selects/deselects

Vectors

Vector Display

Length scale vectors: 0.020

Filled vector arrows

Zero length vectors as point

Draw random vectors

Vector color: Fixed Color

Vector color: Use Data-Layer color

Title: Flow v

Unit: m/s

Number example vectors shown in legend: 5

Map NewMapJeeX

Figure 1: AltstettenLinden
12.00.01 24.07.2022
x/y Cut at k=4 (z=1.3500 m)

Potential Air Temperature

- below 30.73 °C
- 30.73 to 31.14 °C
- 31.14 to 31.56 °C
- 31.56 to 31.97 °C
- 31.97 to 32.38 °C
- 32.38 to 32.79 °C
- 32.79 to 33.20 °C
- 33.20 to 33.61 °C
- 33.61 to 34.02 °C
- above 34.02 °C

Min: 30.32 °C
Max: 34.43 °C

Flow v

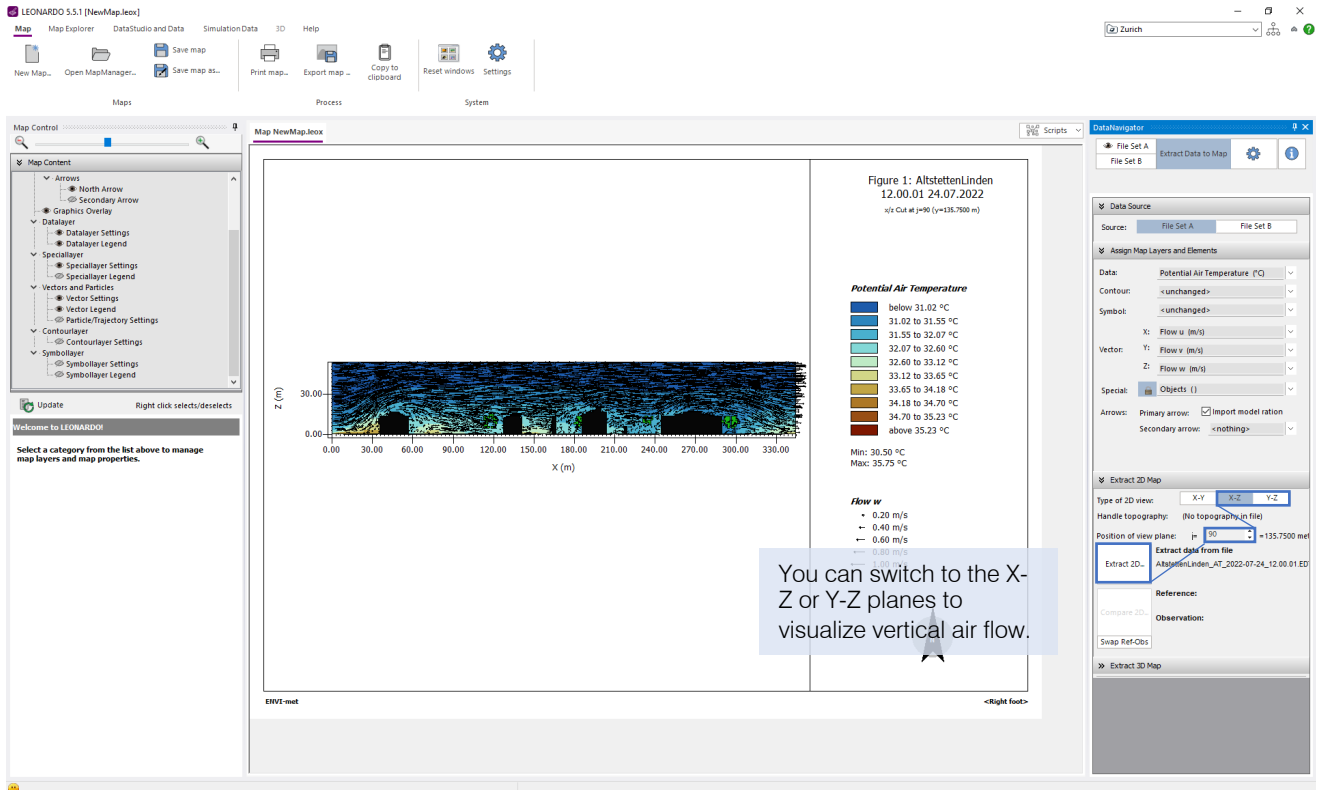
- 0.20 m/s
- 0.40 m/s
- 0.60 m/s
- 0.80 m/s
- 1.00 m/s

Adjust the scale of vectors and the number of random vectors here to enhance the readability of the figure.

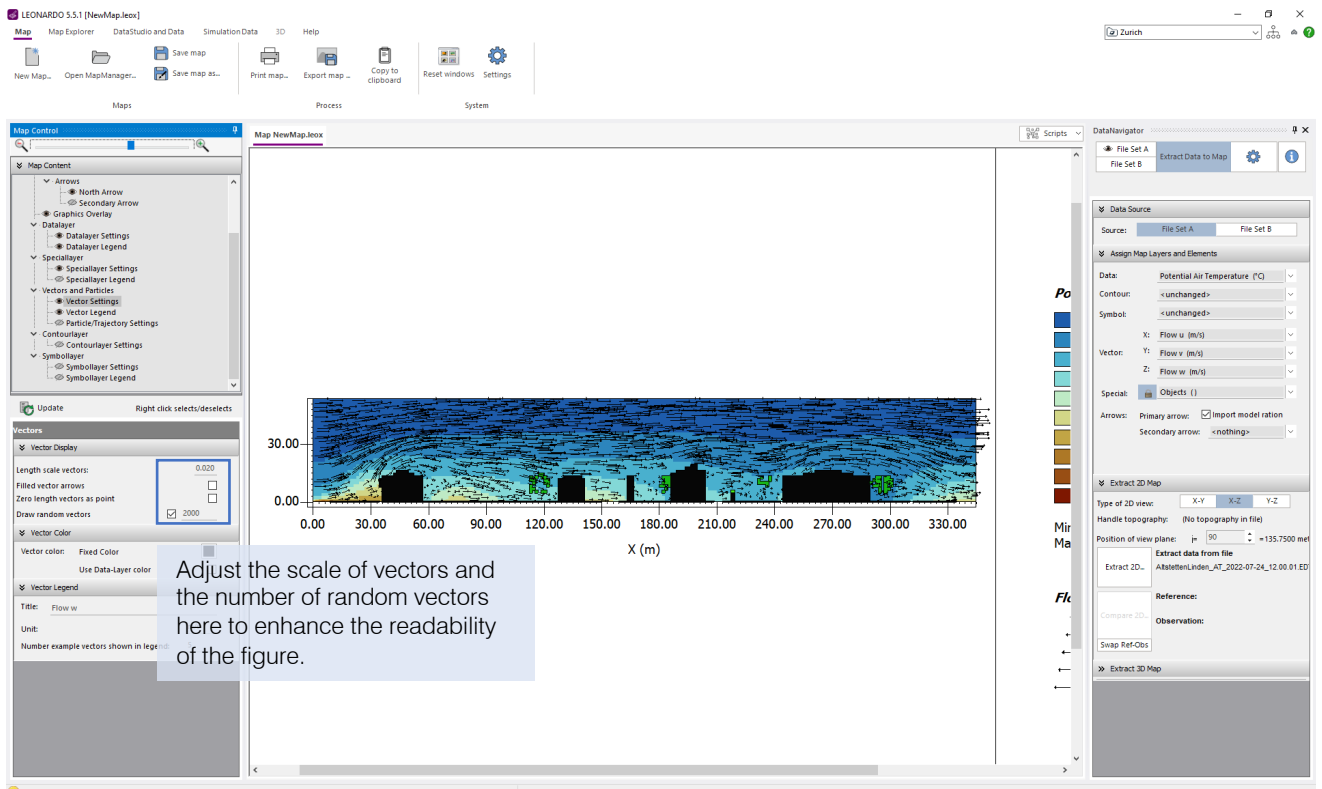
Grid: 91.48 Data: No Data °C Special: 1.00000 (Buildings) Vector: 0.00000,0.00000 m/s Contour: 0.00000 °Place units

Visualization with Leonardo

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Visualization with Leonardo

9

LEONARDO 5.5.1 [NewMap.jscx]

Map Map Explorer DataStudio and Data Simulation Data 3D Help

DataStudio Destination layer Apply Export map layer(s) Export to NetCDF... Import data... Normalize data... Tools

Map Control Map Content

- Arrows
 - North Arrow
 - Secondary Arrow
- Graphics Overlay
 - Datalayer
 - Datalayer Settings
 - Datalayer Legend
 - Speciallayer
 - Speciallayer Settings
 - Speciallayer Legend
 - Vectors and Particles
 - Vector Settings
 - Vector Legend
 - Particle/Trajectory Settings
 - Contourlayer
 - Contourlayer Settings
 - Symbollayer
 - Symbollayer Settings
 - Symbollayer Legend

Update Right click selects/defaults

Contourlayer / Isoline

Isoline Interval

Drawing range: -999.0 to 34.43 Suggest values...

Drawing interval: 51.670

☐ Do not draw lines to special layer ID= 1

Line Style and Colors

Positive isolines: 1 pt

Zero isolines: 1 pt

Negative isoline 1 pt

Use settings from Datalayer legend for color ☐

Labels

Show labels at isolines ☒

Label Distance in grids: 10

Unit name: <Place> Nr of digits: 2

Draw negative labels italic ☒

Label color is same color as isoline ☐

Restrict labels: to medium smooth segments

Figure 1: AltstettenLinden
12.00.01 24.07.2022
x/y Cut at k=4 (=1.3500 m)

Wind Speed

< 0.00 m/s
0.10 m/s
0.20 m/s
0.30 m/s
0.40 m/s
0.50 m/s
0.60 m/s
0.70 m/s
0.80 m/s
0.90 m/s
Min: 0.00 m/s
Max: 0.99 m/s

Flow v

0.20 m/s
0.40 m/s

Right-click on a grid point on the map, then click on 'Explore...'

Explore grid is 108, j=114 [XY-Cut k=4]

Explore map grid 108, 114...

Set upper left marker here

Set lower right marker here

Clear markers

Cut map to window...

Cut map to window (Animate)...

You can also visualize wind and north direction

Data Navigator

File Set A File Set B Extract Data to Map

Data Source

Sources: File Set A File Set B

Assign Map Layers and Elements

Data: Wind Speed (m/s)

Contour: Potential Air Temperature (°C)

Symbol: <unchanged>

X: Flow u (m/s)

Y: Flow v (m/s)

Z: Flow w (m/s)

Special: Objects (1)

Arrows: Primary arrow: ☒ Import model ration Secondary arrow: Wind direction

Extract 2D Map

Type of 2D view: X-Y X-Z Y-Z

Handle topography: (No topography in file)

Position of view plane: k= 4 =1.3500 meter

Extract 2D: Extract data from file AltstettenLinden_AT_2022-07-24_12.00.01 ED

Reference: Observation:

Swap Ref-Obs

Extract 3D Map

Grid: 108,114 Data: 0.57437 m/s Special: 0.00000 (???) Vector: 0.24877,0.51767 m/s Contour: 32.58809 <Place unit>

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LEONARDO 5.5.1 [NewMap.jscx]

Map Map Explorer DataStudio and Data Simulation Data 3D Help

DataStudio Destination layer Apply Export map layer(s) Export to NetCDF... Import data... Normalize data... Tools

Map Control Map Content

- Arrows
 - North Arrow
 - Secondary Arrow
- Graphics Overlay
 - Datalayer
 - Datalayer Settings
 - Datalayer Legend
 - Speciallayer
 - Speciallayer Settings
 - Speciallayer Legend
 - Vectors and Particles
 - Vector Settings
 - Vector Legend
 - Particle/Trajectory Settings
 - Contourlayer
 - Contourlayer Settings
 - Symbollayer
 - Symbollayer Settings
 - Symbollayer Legend

Update Right click selects/defaults

Contourlayer / Isoline

Isoline Interval

Drawing range: -999.0 to 34.43 Suggest values...

Drawing interval: 51.670

☐ Do not draw lines to special layer ID= 1

Line Style and Colors

Positive isolines: 1 pt

Zero isolines: 1 pt

Negative isoline 1 pt

Use settings from Datalayer legend for color ☐

Labels

Show labels at isolines ☒

Label Distance in grids: 10

Unit name: <Place> Nr of digits: 2

Draw negative labels italic ☒

Label color is same color as isoline ☐

Restrict labels: to medium smooth segments

Figure 1: AltstettenLinden

Map Explorer

Exploring map element: 113,119,4 XY-Cut at k=4

Data View Name: AltstettenLinden_AT_113_119

Chart

Data Sources

24.07.2022

00:00 02:00 04:00 06:00 08:00 10:00

12:00 14:00 16:00 18:00 20:00

25.07.2022

00:00 02:00 04:00 06:00 08:00 10:00

12:00 14:00 16:00 18:00 20:00

Set time series to whole range

Set Time Series Start...

Set Time Series End...

Select Variables

Objects (1)

Flow u (m/s)

Flow v (m/s)

Flow w (m/s)

Wind Speed (m/s)

Wind Speed Change (%)

Wind Direction (deg)

Pressure Perturbation (dPa)

Potential Air Temperature (K)

Air Temperature Delta (K)

Air Temperature Change (K)

Specific Humidity (g/kg)

Relative Humidity (%)

TXE (m²/m²)

TXE Dissipation (m²/m²)

(Select a variable above)

Color: Topo4

Point Style

Axis: Left Axis Right Axis

Data displayed in map: 24.07.2022 12:00:01

Export Sheets... Send to DataStudio...

Right-click on a point within the time span, then set the time series to cover the whole range.

Data Navigator

File Set A File Set B Extract Data to Map

Data Source

Sources: File Set A File Set B

Assign Map Layers and Elements

Data: Wind Speed (m/s)

Contour: Potential Air Temperature (°C)

Symbol: <unchanged>

X: Flow u (m/s)

Y: Flow v (m/s)

Z: Flow w (m/s)

Special: Objects (1)

Arrows: Primary arrow: ☒ Import model ration Secondary arrow: Wind direction

Extract 2D Map

Type of 2D view: X-Y X-Z Y-Z

Handle topography: (No topography in file)

Position of view plane: k= 4 =1.3500 meter

Extract 2D: Extract data from file AltstettenLinden_AT_2022-07-24_12.00.01 ED

Reference: Observation:

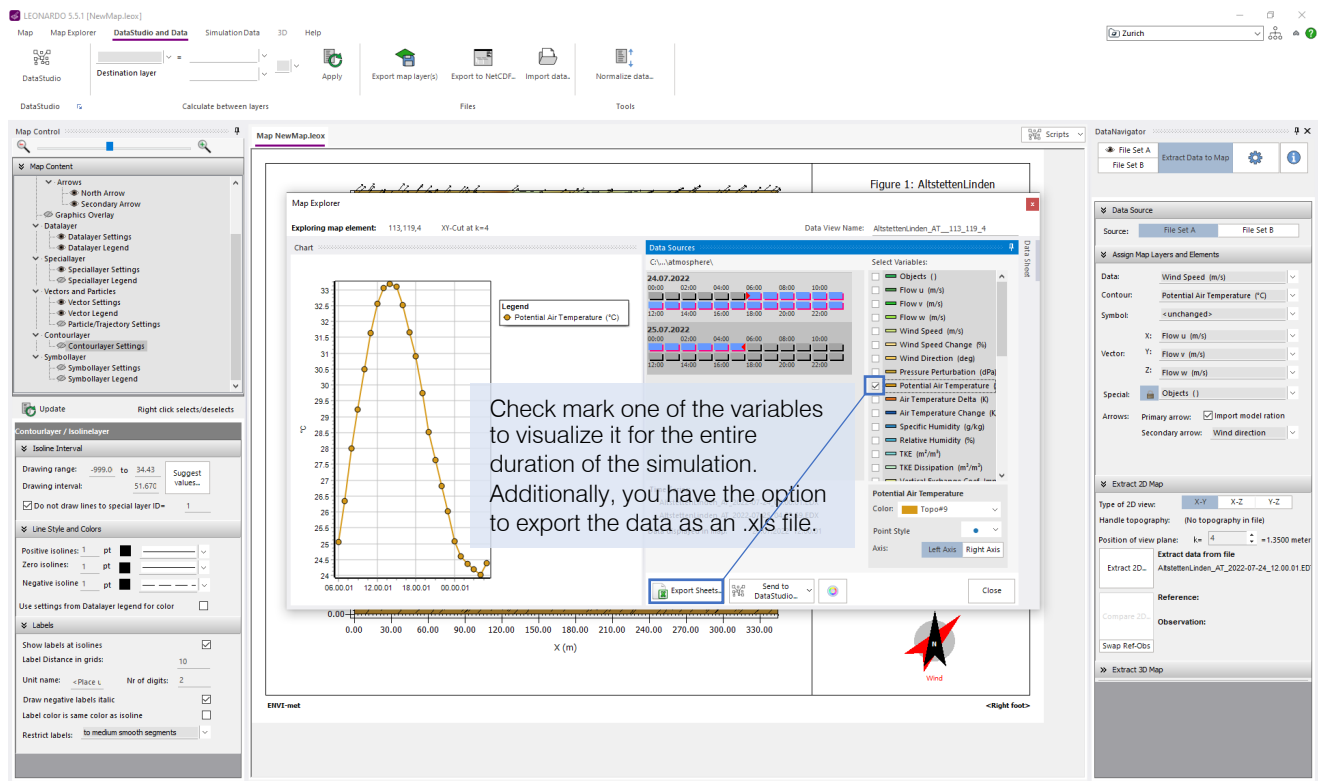
Swap Ref-Obs

Extract 3D Map

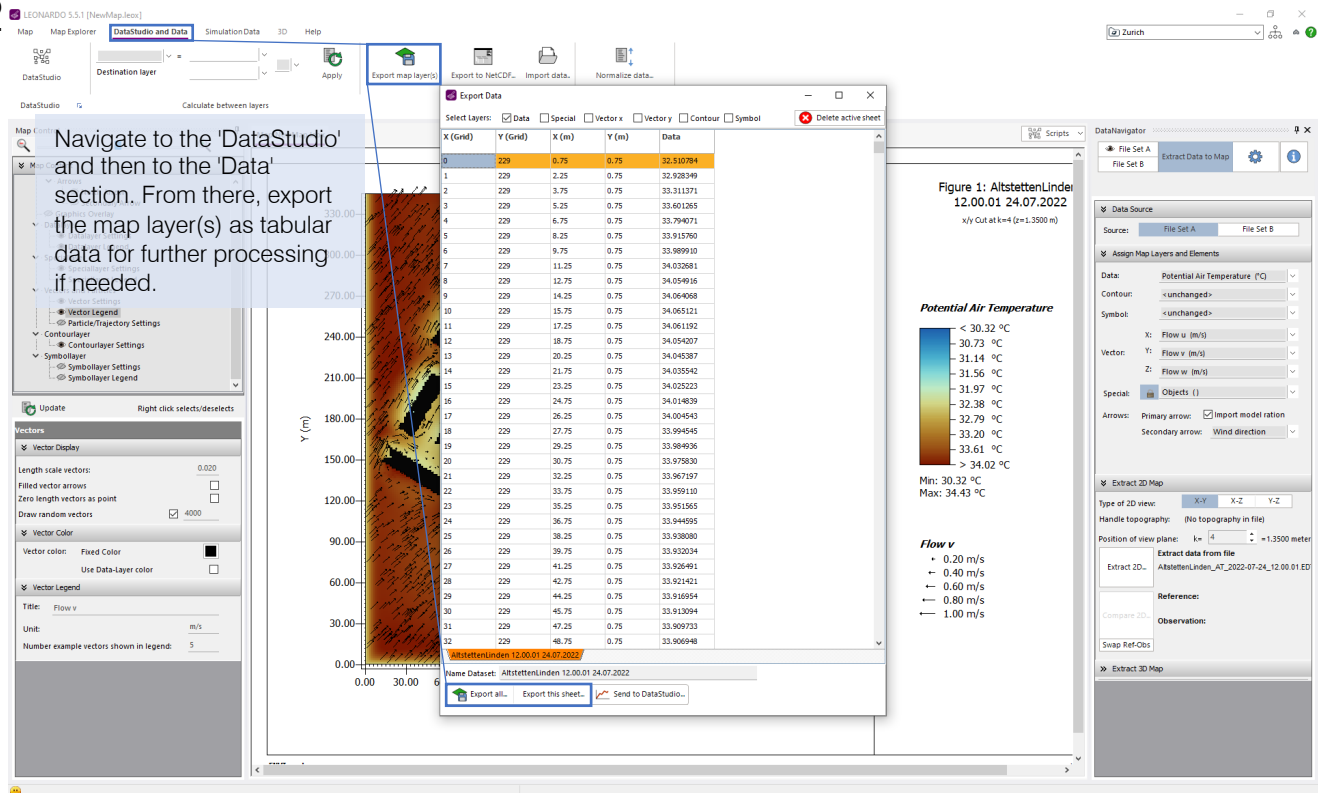
Grid: 108,114 Data: 0.57437 m/s Special: 0.00000 (???) Vector: 0.24877,0.51767 m/s Contour: 32.58809 <Place unit>

Visualization with Leonardo

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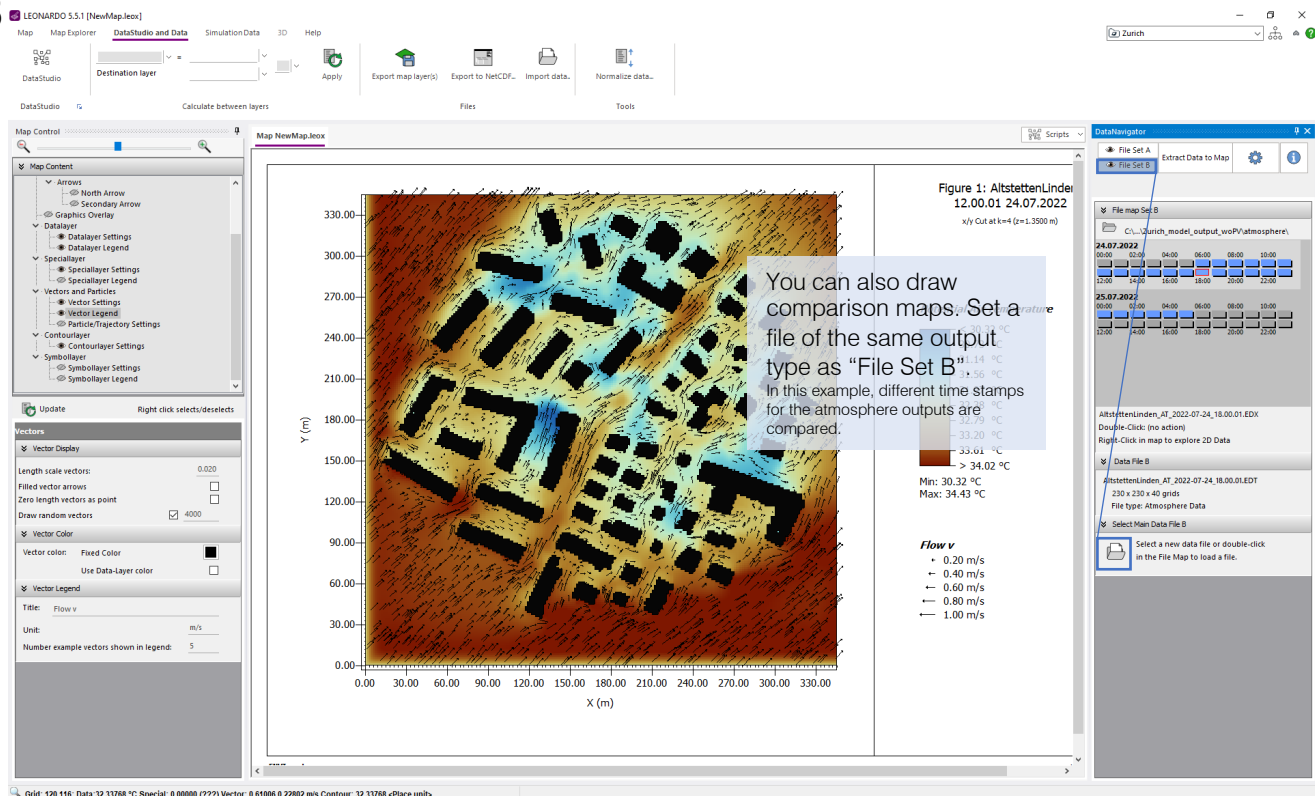


12

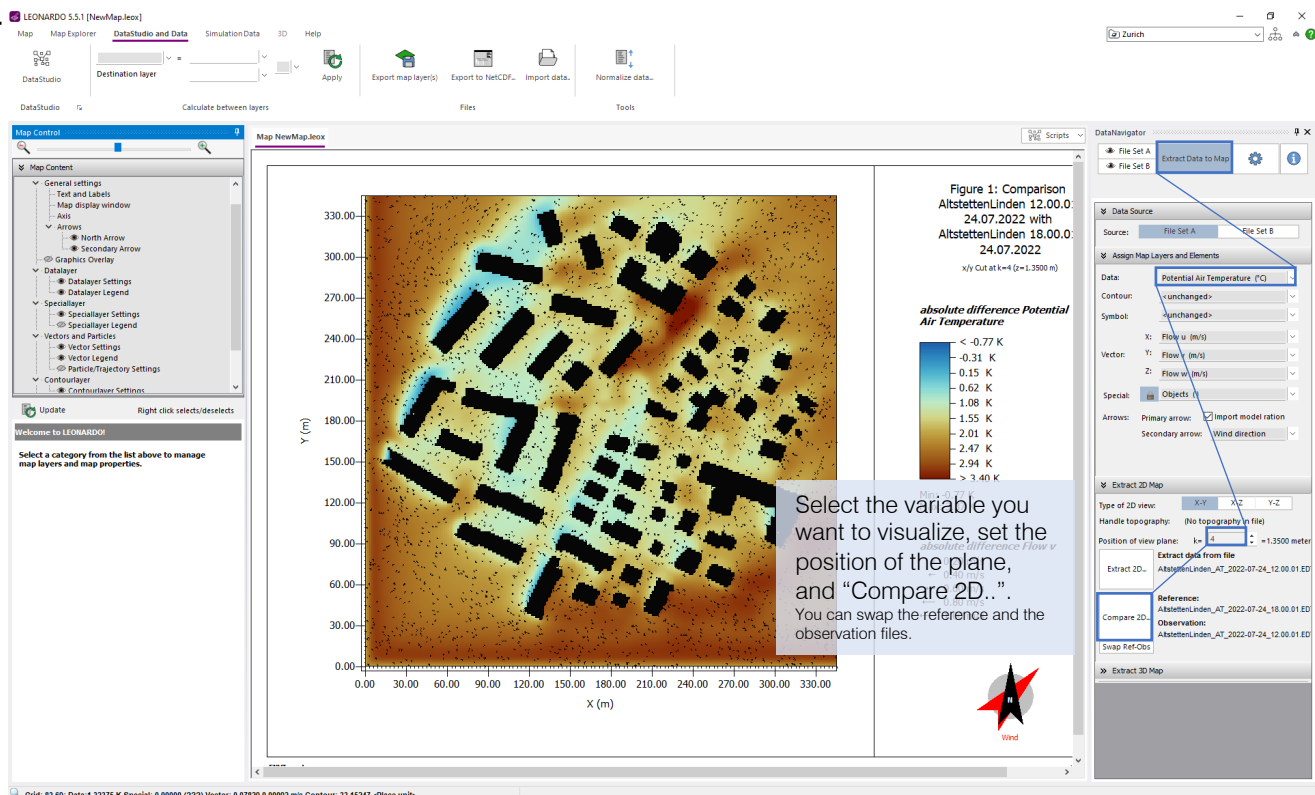


Visualization with Leonardo

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Visualization with Leonardo

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Map Control

Map Content

- Arrows
- Secondary Arrow
- Graphics Overlay
- Datayer
- Datayer Settings
- Speciallayer
- Speciallayer Settings
- Speciallayer Legend
- Vectors and Particles
- Vector Settings
- Particle/Trajectory Settings
- Contourlayer
- Contourlayer Settings
- Symbolayer
- Symbolayer Settings
- Symbolayer Legend

Update

Right click selects/deselects

Contourlayer / Isolines

Isoline Interval

Drawing range: -999.0 to 34.43

Drawing interval: 51.67%

Do not draw lines to special layer ID: 1

Line Style and Colors

Positive isolines: 1 pt

Zero isolines: 1 pt

Negative isoline: 1 pt

Use settings from Datayer legend for color

Labels

Show labels at isolines

Label Distance in grids: 10

Unit name: <Place> Nr of digits: 2

Draw negative labels italic

Label color is same color as isoline

Restrict labels: to medium smooth segments

Figure 1: AltstettenLinden 12.00.01 24.07.2022

x/y Cut at k=4 (z=1.3500 m)

Wind Speed

- < 0.00 m/s
- 0.10 m/s
- 0.20 m/s
- 0.30 m/s
- 0.40 m/s
- 0.50 m/s
- 0.60 m/s
- 0.80 m/s
- 1.00 m/s

SWR-mat

Wind

Right foot

DataNavigator

File Set A

File Set B

File map Set A

24.07.2022

00:00 02:00 04:00 06:00 08:00 10:00

12:00 14:00 16:00 18:00 20:00 22:00

25.07.2022

00:00 02:00 04:00 06:00 08:00 10:00

12:00 14:00 16:00 18:00 20:00 22:00

AltstettenLinden, BLDG_2022-07-24_14.00.01.EDX

Double-Click: Extract 3D cut

Right-Click in map to explore 2D Data

Data File A

AltstettenLinden, BLDG_2022-07-24_14.00.01.EDT

230 x 230 x 40 grids

File type: Facade Data

Select main Data File A

Select a new data file or double-click in the File Map to load a file.

Grid: 105.79; Data: 0.04135 m/s Special: 0.00000 (???) Vector: 0.02025, 0.03128 m/s Contour: 32.48251 <Place units>

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Map Control

Map Content

- Text and Labels
- 3D Settings
- Light&Vision
- Image Overlay
- Datayer3D
- Datayer3D Settings
- Datayer3D Legend
- Speciallayer3D
- Speciallayer3D Settings
- Speciallayer3D Legend
- Vectors3D
- Vector3D Settings
- Particle/Trajectory3D Settings
- TubularRef3D Settings
- Isosurface3D Settings
- Isosurface3D Legend
- Facadelay3D
- Facadelay3D Settings
- Facadelay3D Legend

Update

Right click selects/deselects

Facadelay3D legend

Color Mapping

Default

Start colorizing at index: 10

Legend Settings

Title: Wall System: SW Rad. received

Unit name: W/m2 Number of digits: 2

Other fields in legend: Min/Max No Data

Legend Classes

Number colors/classes: 10

Step size/Class width: 94.24

First value in legend: 0.002

No Data ID: -999

Figure 1: AltstettenLinden 14.00.01 24.07.2022

x/y Cut at k=4 (z=1.3500 m)

Wall System: SW Rad. received

W/m2

- 0.00 W/m2
- 629.72 W/m2
- 753.96 W/m2
- 848.21 W/m2
- 942.45 W/m2

Min: 0.00 W/m2

Max: 942.45 W/m2

DataNavigator

File Set A

File Set B

Data Source

Source: File Set A File Set B

Assign Map Layers and Elements

Data: <unchanged>

iso-3D: <unchanged>

Vector: X: <unchanged> Y: <unchanged> Z: <unchanged>

Spectat: Objects (from Facade File)

Arrows: Primary arrow: Import model ratio Secondary arrow: Wind direction

Facades (3D): Wall System: SW Rad. received

Extract 3D Map

Extract 3D Facade

Extract data from file AltstettenLinden_BLDG_2022-07-24_14.00.01

References: Observation

Swap Ref - Obs

Make sure that the Facadelay3D settings and legend are turned on. You can select "floating" for smoother visualization of colors.

Move to the "Extract Data to Map" page. Open the "Extract 3D Map" section. Select one of the variables from "Facades (3D)" and "Extract 3D Façade."

Visualization with Leonardo

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LEONARDO 5.5.1 [NewMap.jmx]

Map Explorer | DataStudio and Data | Simulation Data | 3D | Help

Map Control | Map NewMap.jmx

Map Content

- Text and Labels
- 3D Settings
- Light & Vision
- Image Overlay
- Datalayer3D
 - Datalayer3D Settings
 - Datalayer3D Legend
- Speciallayer3D
 - Speciallayer3D Settings
 - Speciallayer3D Legend
- Vectorlayer3D
 - Vector3D Settings
 - Particle/Trajectory3D Settings
 - TubularRef3D Settings
- Isosurface3D
 - Isosurface3D Settings
- Facade3D
 - Facade3D Settings
 - Facade3D Legend

Speciallayer3D Settings

Code: 12
Label: Vegetation: LAD 0.5 - 1.0
Type: Item

Special Coding and Display Style

Color: Filestyle:
Line width: 1 pt
Line color:

Drawing Style 2D Maps

Color: Transparency:
Draw cube outlines
Line width: 1 pt
Line color:

Figure 1: AltstettenUnder
14.00.01 24.07.2022
x/y Cut at k=4 (z=1.3500 m)

Wall System: SW Rad. received

Min: 0.00 W/m²
Max: 942.45 W/m²

Data Navigator

Source: File Set A | File Set B

Assign Map Layers and Elements

Data:
Iso-3D:
X:
Y:
Z:
Special:
Arrows: Primary arrow: ☐ Import model ration
Secondary arrow: Wind direction
Facades (3D): Wall System: SW Rad. received

Extract 2D Map

Extract data from file
AltstettenUnder_BLDG_2022-07-24_14.00.01

Extract 3D Map

Extract 3D Facade...
Compare 3D Facade...
Reference:
Observation:
Swap Ref - Obs

You can change the visibility of objects on the map. Double-click on the object category to adjust its display settings in the pop-up menu.

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LEONARDO 5.5.1 [NewMap.jmx]

Map Explorer | DataStudio and Data | Simulation Data | 3D | Help

Map Control | Map NewMap.jmx

Map Content

- Text and Labels
- 3D Settings
- Light & Vision
- Image Overlay
- Datalayer3D
 - Datalayer3D Settings
 - Datalayer3D Legend
- Speciallayer3D
 - Speciallayer3D Settings
 - Speciallayer3D Legend
- Vectorlayer3D
 - Vector3D Settings
 - Particle/Trajectory3D Settings
 - TubularRef3D Settings
- Isosurface3D
 - Isosurface3D Settings
- Facade3D
 - Facade3D Settings
 - Facade3D Legend

Speciallayer3D Settings

Code: 12
Label: Vegetation: LAD 0.5 - 1.0
Type: Item

Special Coding and Display Style

Color: Filestyle:
Line width: 1 pt
Line color:

Drawing Style 2D Maps

Color: Transparency:
Draw cube outlines
Line width: 1 pt
Line color:

Figure 1: AltstettenUnder
14.00.01 24.07.2022
x/y Cut at k=4 (z=1.3500 m)

Wall System: SW Rad. received

Min: 0.00 W/m²
Max: 942.45 W/m²

Data Navigator

Source: File Set A | File Set B

Assign Map Layers and Elements

Data:
Iso-3D:
X:
Y:
Z:
Special:
Arrows: Primary arrow: ☐ Import model ration
Secondary arrow: Wind direction
Facades (3D): Wall System: SW Rad. received

Extract 2D Map

Extract data from file
AltstettenUnder_BLDG_2022-07-24_14.00.01

Extract 3D Map

Extract 3D Facade...
Compare 3D Facade...
Reference:
Observation:
Swap Ref - Obs

You can navigate using the view control pop-up.

Zoom in and right-click on a grid on the façade surface.

Visualization with Leonardo

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The screenshot shows the Leonardo 5.5.1 [NewMap] interface. The main window displays a 2D facade visualization of a building facade. A data source is selected, and a chart shows the facade element's performance over time. A text box indicates: "Right-click and select the entire time span. Check the variables you want to visualize, and then export the data into an .xls file." The interface includes a Map Explorer, Map Control, and Data Navigator panels.

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The screenshot shows the Leonardo 5.5.1 [NewMap] interface. The main window displays a 3D facade visualization of a building facade. A data source is selected, and a chart shows the facade element's performance over time. A text box indicates: "You can similarly make comparisons in 3D from the extract 3D Map section. Upload two façade files." The interface includes a Map Explorer, Map Control, and Data Navigator panels.

Visualization with Leonardo

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